

Sensing the world

In the near future, intelligent little sensors are all set to network themselves into a useful web around you

They could be as big as pocket radios, or as small as household dust—tomorrow's biggest pervasive networks are going to comprise tiny sensors, sampling everything on Earth and beyond. The idea is simple, yet startlingly effective—flood the world with a vast number of intelligent network sensors and use them to observe, study and analyse everything around them. That's what the incubators of tomorrow's technology, such as National Aeronautics & Space Administration (NASA), the Palo Alto Research Centre (PARC) and several universities have been working on—wireless sensor networks.

Sensors are all over our world—right from tiny ones that monitor your oven, to those that monitor a

car's internal mechanisms. Sensors are also wired into smoke and fire detectors that cover industrial buildings and households. They have a simple job—to sense changes in physical conditions and act on pre-programmed instructions, or relay back information to a central authority. Thus, their behaviour is very predictable.

Going web-ing

That's where things are changing. At NASA's Jet Propulsion Labs (JPL)(<http://sensor-webs.jpl.nasa.gov/>), located within the California Institute of Technology, a team of dedicated scientists headed by Kevin A. Delin, is working towards teaming up simple sensors into a gigantic web of sorts. They have christened it the Sensor Webs project, and are looking at the emergence of intelligent, networked sensors, and focusing on acquiring and interpreting physical information over a large spatial area.

The Sensor Webs project is working with fixed, as well as mobile sensor platforms, which are made up of tiny multiple sensors that apart from sensing changes in the environment, can also communicate with each other. However, instead of using few expensive sensors

that read and transmit high-resolution physical data such as readings taken over a wide area, the project looks at multiple, low-cost sensors working together with more sensors sampling, analysing and sharing local physical information. Thus, they form a wireless web of sensors that read, share and analyse external conditions, and go even further by reacting and adapting to the environment.

Each of the sensors has a processor, memory, radio transmission capabilities, a battery cell, as well as a solar cell—all of which sits within a plastic body. Each pod holds several such small sensors that first sample the physical characteristics they were designed for, share this information with neighbouring pods, and analyse and alter their behaviour accordingly. The information that is read and processed, hops all along the pod and other pods, till a common point is found from where the data can be shared with a central computing environment, called the base station. Redundancy is built into the web; if one sensor happens to go down, the neighbouring sensors realise this, and adjust their sampling rates in order to make up for the loss. In effect, the sensors are smart enough to understand the information, and then uplink that processed knowledge to the computers and teams monitoring them.

Now all this information has to be easily accessible to the team handling and

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